

Review Quiz 7 Math 2210

Ex 1 Show that $\begin{pmatrix} 4 & 1 & 2 \\ 1 & 0 & 3 \\ 2 & 3 & 1 \end{pmatrix}$ is not pos def.

Repeat for $\begin{pmatrix} -1 & 2 \\ 2 & 4 \end{pmatrix}$, $\begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}$, $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, $\begin{pmatrix} 4 & 1 & 2 \\ 1 & 1 & 3 \\ 2 & 3 & 1 \end{pmatrix}$

Ex 2 Show that $\begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix}$ $\begin{pmatrix} 2 & 2 & 0 \\ 2 & 5 & 3 \\ 0 & 3 & 8 \end{pmatrix}$ are pos def.

Ex 3 For what c is $\begin{bmatrix} c & 1 & 1 \\ 1 & c & 1 \\ 1 & 1 & c \end{bmatrix}$ pos def.

Ex 4 Compute $A^{1/2}$ where A is as in Ex 2

Ex 5 Describe the curve $(x,y) A \begin{pmatrix} x \\ y \end{pmatrix} = 1$ where $A = \begin{pmatrix} -1 & 2 \\ 2 & 4 \end{pmatrix}$

$A = \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}$ $A = \begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix}$ $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ (What shape is it.
What are the axes?)

Ex 6 Define similarity of matrices.

Show that similarity is an equivalence relation.

Show that similar matrices have the same eigenvalues. How are the eigenvectors related?